ENHANCED LSTM TIME SERIES FORECASTING

Experimental Results and Model Evaluation

MODEL PERFORMANCE COMPARISON

Model	MSE	MAE	RMSE
Baseline LSTM	3.452	1.4252	1.858
Improved LSTM + Attention	3.7545	1.4979	1.9376
Transformer	3.7411	1.519	1.9342

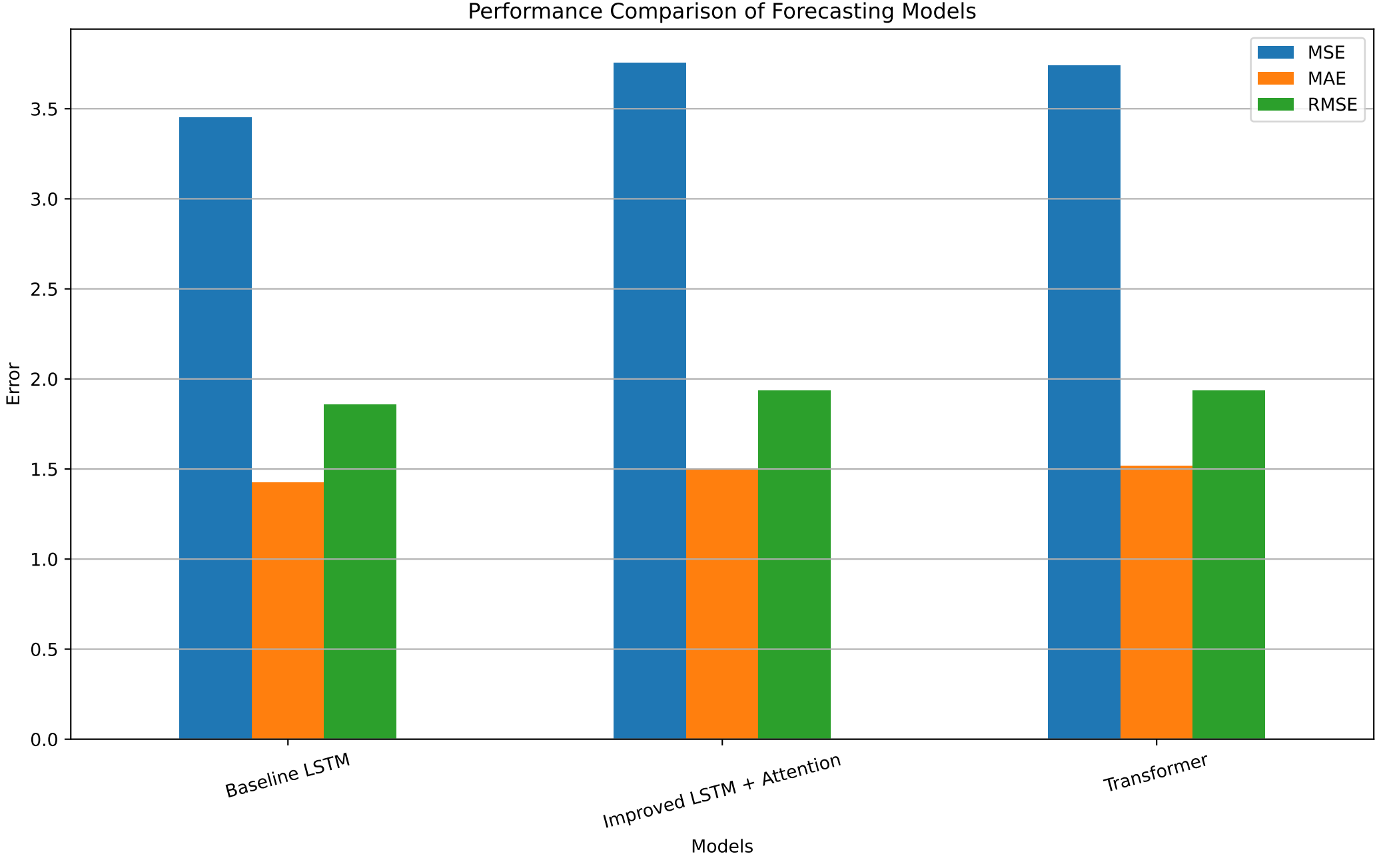
Model: Baseline LSTM

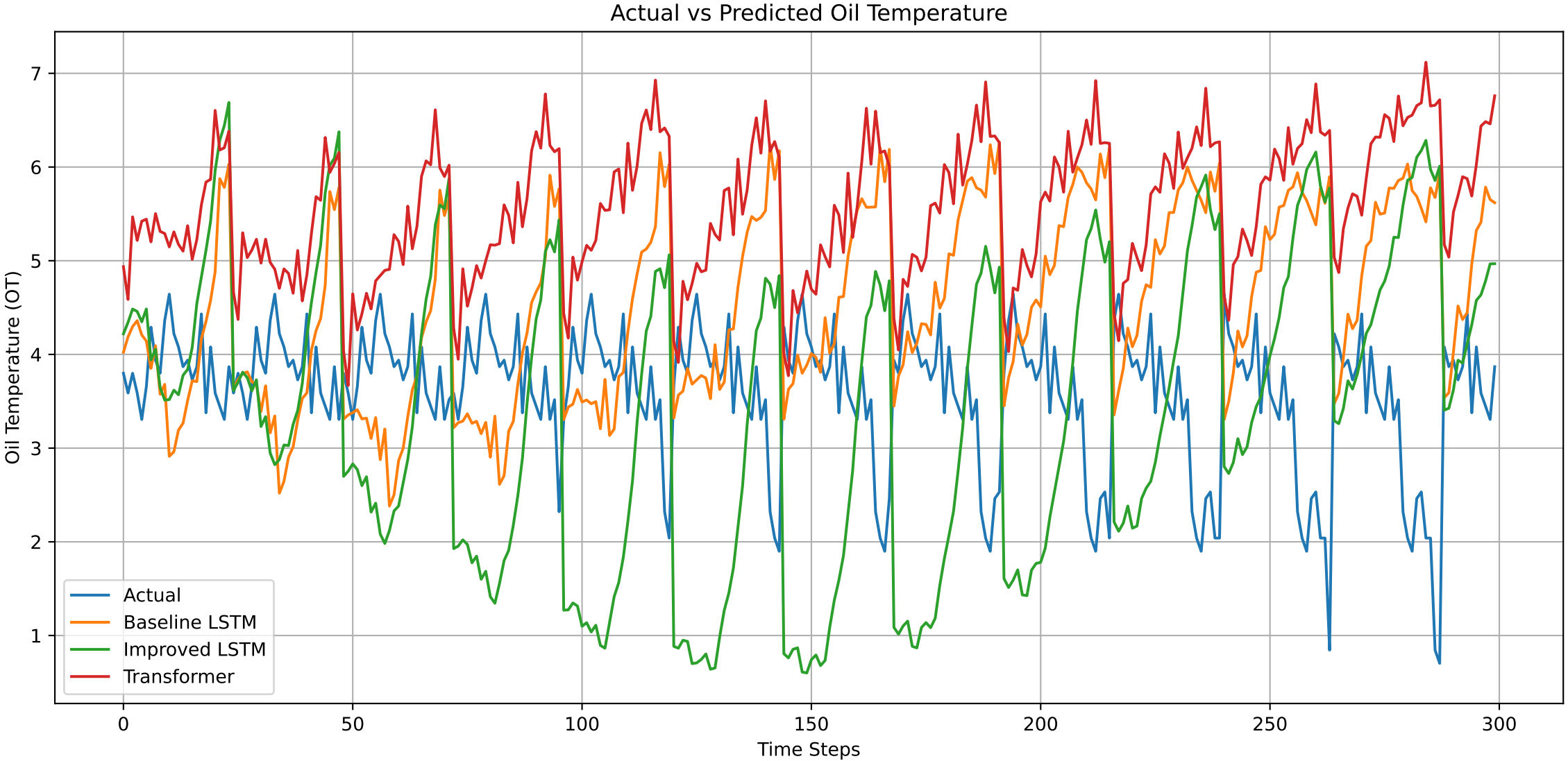
RMSE: 1.8580

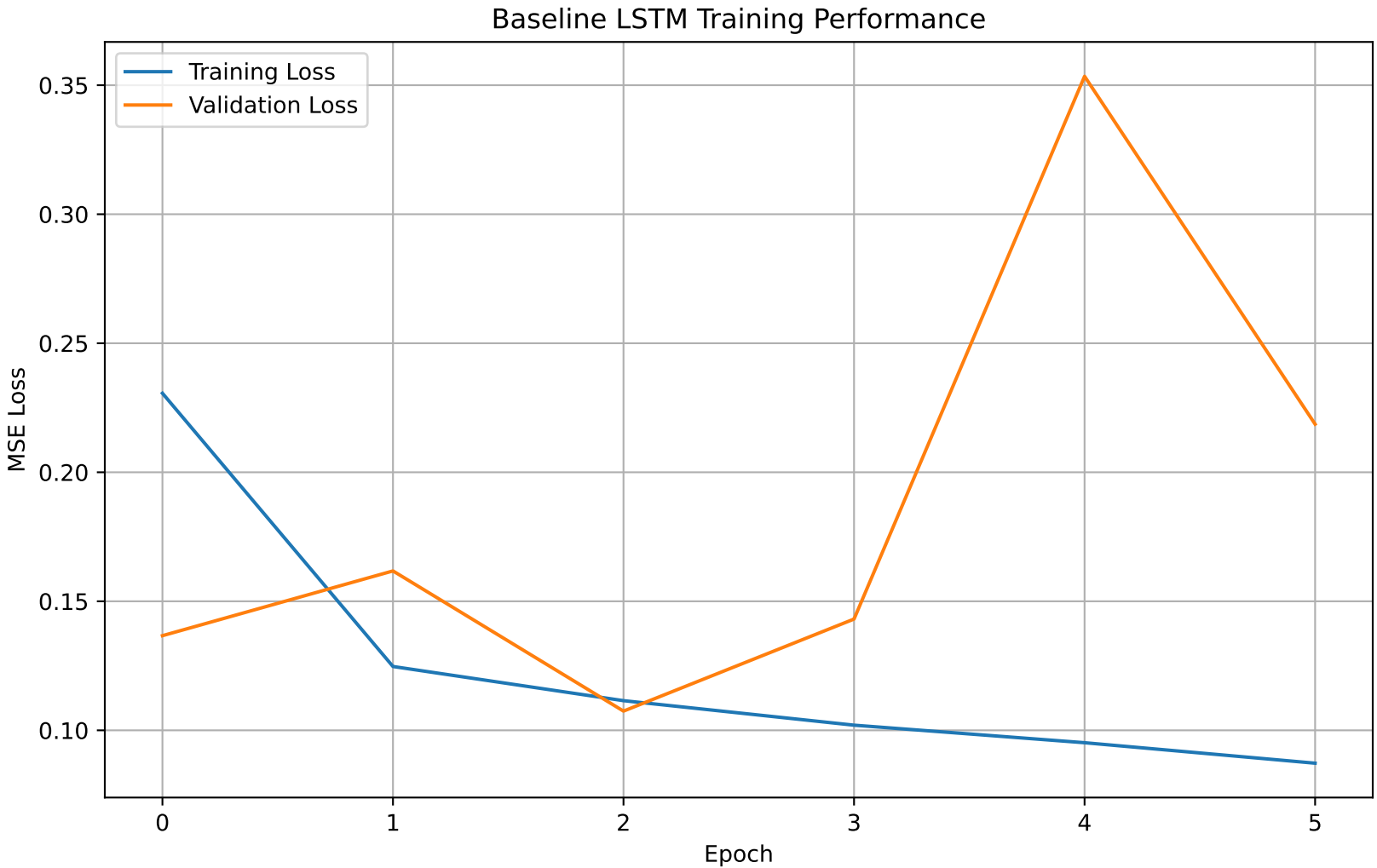
BEST PERFORMING MODEL

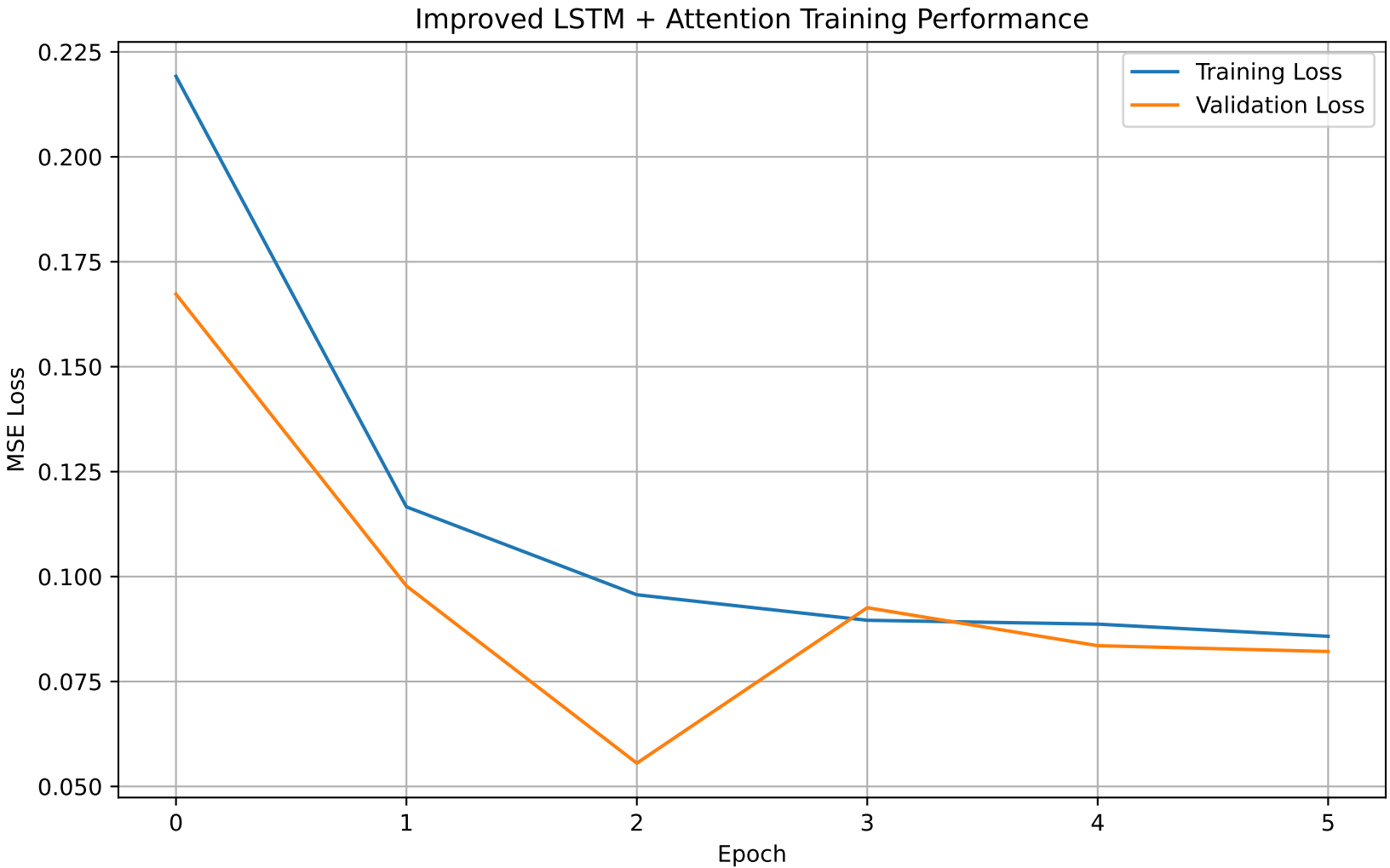
MAE: 1.4252

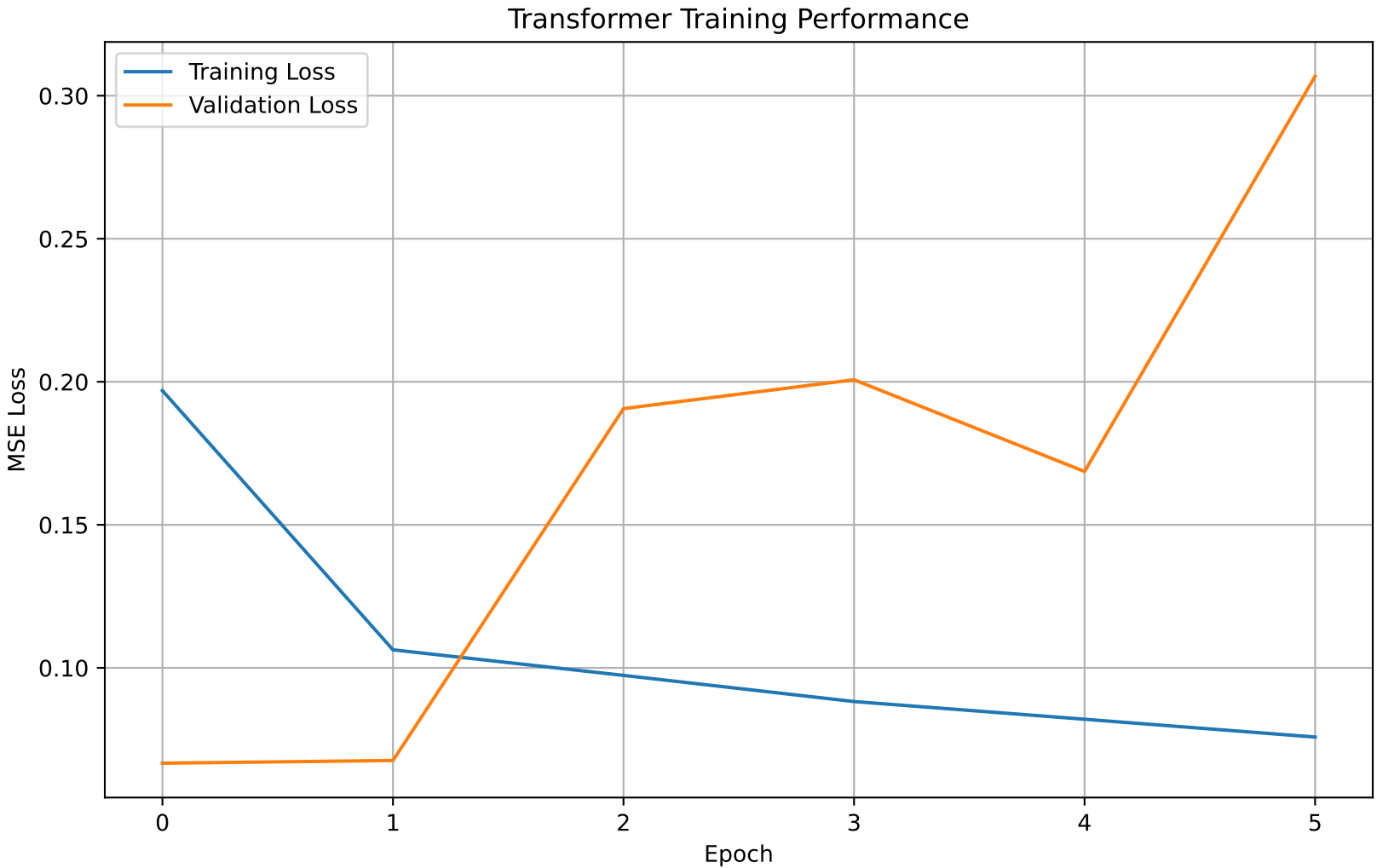
MSE: 3.4520


Performance Comparison of Forecasting Models
MSE
MAE
RMSE
Error
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Baseline LSTM
Improved LSTM + Attention
Transformer
Models


Actual vs Predicted Oil Temperature
Oil Temperature (OT)
Time Steps
Actual
Baseline LSTM
Improved LSTM
Transformer
1
2
3
4
5
6
7
0
50
100
150
200
250
300


Baseline LSTM Training Performance
Training Loss
Validation Loss
MSE Loss
0.35
0.30
0.25
0.20
0.15
0.10
0
1
2
3
4
5
Epoch


Improved LSTM + Attention Training Performance
Training Loss
Validation Loss
MSE Loss
Epoch
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0
1
2
3
4
5


Transformer Training Performance
Training Loss
Validation Loss
MSE Loss
0.30
0.25
0.20
0.15
0.10
0
1
2
3
4
5
Epoch

Actual and Predicted Values (First 30 Samples)

Actual	Baseline LSTM	Improved LSTM	Transformer
3.799	4.022	4.219	4.936
3.588	4.187	4.342	4.587
3.799	4.297	4.485	5.468
3.588	4.361	4.458	5.216
3.306	4.207	4.348	5.422
3.658	4.143	4.485	5.443
4.291	3.851	3.942	5.201
3.939	4.093	4.056	5.503
3.799	3.575	3.646	5.313
4.362	3.68	3.513	5.291
4.643	2.913	3.518	5.148
4.221	2.961	3.618	5.308
4.08	3.193	3.57	5.175
3.869	3.266	3.778	5.102
3.939	3.508	3.84	5.374
3.728	3.719	4.075	5.014
3.869	3.71	4.543	5.226
4.432	4.168	4.829	5.597
3.377	4.351	5.111	5.84
4.08	4.576	5.419	5.868
3.588	4.883	5.963	6.604
3.447	5.876	6.284	6.182
3.306	5.782	6.443	6.205
3.869	6.031	6.689	6.382
3.588	3.623	3.614	4.663
3.799	3.734	3.704	4.373
3.588	3.795	3.811	5.299
3.306	3.814	3.752	5.033
3.658	3.709	3.611	5.114
4.291	3.677	3.729	5.229